

Explainable AI for Post-COVID Psychological Profile Prediction

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ABSTRACT

The psychological profile prediction of post-COVID-19 patients presents a dynamic and complex challenge, raising questions about how machine learning models produce mental health classifications. Many AI systems, particularly deep learning algorithms, function as "black boxes," making their decision-making process difficult to understand despite their excellent performance. By applying explainability techniques, we can uncover the inner workings of these models and gain insight into how their classifications are generated. Here we show that SHAP (SHapley Additive exPlanations) values can be utilized to interpret model decisions and produce counterfactual explanations for diagnoses. These counterfactuals aim to provide deeper insight into AI-generated predictions of post-COVID psychological profiles, enhancing transparency and fostering trust in these systems.

1. INTRODUCTION

A study by Čosić et al. (2021) reveals that an AI-assisted psychotherapeutic approach may assist in lowering costs and providing better and faster classification of ex-COVID-19 patients according to high, medium, or low risk of chronic psychopathology. However, this approach utilizes methods based on supervised machine learning (ML) that sacrifices model explainability in exchange for maximizing classification accuracy. Despite the excellent performance of these models, the decision-making process is often opaque. Explainability is a key component for reliable and trustworthy predictions, personalized treatments, and appropriate preventive suggestions for mental health deterioration in ex-COVID-19 patients. This leads to efforts in building explainable AI as a prerequisite for deploying critical applications (Singh et al., 2024).

SHapley Additive exPlanations (SHAP) utilizes game theory to explain the results of black-box models. It is based on classical Shapley value estimation, which is a regression method that computes the weighted average of all possible differences. By computing SHAP values, we are able to point out to each feature the change in the expected model prediction when conditioning on those features (Lundberg & Lee, 2017).

Other explainable method include Local Interpretable Model-agnostic Explanations (LIME) which generates local approximations of model predictions (Ribeiro et al., 2016), and Deep Learning Important Features (DeepLIFT) which decomposes the output prediction of a neural network by back-propagating contributions of all neurons in the network to each input feature (Shrikumar et al., 2019).

Additionally, Counterfactual SHAP (CF-SHAP) is a modular framework that leverages the fitness of mathematical properties of Shapley values, and by extension SHAP values, with human intuition for feature importance in order to generate counterfactual explanations (Albini et al., 2022).

Since the computation of exact SHAP values are costly, especially in large aggregates, approximations have to be made in order to explain the classification model.

For this study, we propose utilizing the Shapley kernel to allow us to start from a broad, model-agnostic solution (Lundberg & Lee, 2017). By adapting Kshetry & Kantardzic's (2024) XAI framework, we will be able to leverage these SHAP value estimates into counterfactuals that allow us to build contextualized and interpretable explanations for causal understanding.

2. METHODOLOGY

We begin with preprocessing the data to build our baseline model using PyTorch. We will use the sociodemographic, COVID-19-related, mental health, and coping behavior information columns from our dataset to predict our target features which are the mental health metrics. We will then utilize state-of-the-art XAI methods to implement our Explanation Framework. Generated results from our baseline model and explanation framework will be evaluated by domain experts in the field of psychology. In building our explanation framework, we will utilize InterpretML and DiCE for explainability and counterfactual explanations, alongside BioMistral, which will aid in packaging our counterfactual explanations.

2.1 Data Collection and Preparation

For this study, we refer to a dataset from an online survey conducted on the Japanese population when the number of

COVID-19-positive cases was high and when the COVID-19 pandemic was officially declared to be over. The dataset has 92 attributes, including socio-demographic, COVID-19-related, mental health, and coping behavior information from 2,659 respondents (Nuñez, 2023). This dataset contains six surveys, spanning from 2020 to 2023.

After conducting a Wilcoxon signed-rank statistical test on the following mental health metrics: K6 for emotional distress, PHQ-9 for depression, GAD-7 for anxiety, SSS-8 for somatic symptom burden, and the UCLA loneliness scale.

For these tests, we set our null hypothesis as: "There is no significant difference in these mental health metrics during the COVID-19 pandemic and after it." Our alternative hypothesis (for the one-sided test) was: "There is a statistically significant increase in these mental health metrics after COVID-19."

After conducting the Wilcoxon signed-rank test, we found that each metric had a p-value of less than 0.05, allowing us to reject the null hypothesis and infer that there is a statistically significant increase in these mental health metrics after COVID-19—which is concerning.

2.2 Feature Selection

Data from the fifth survey was used for the features. There are four main categories in the dataset attributes: Socio-Demographic Information, COVID-19-Related Information, Mental Health Information, and Coping Strategies Information. Only particular attributes per category were chosen in order to loosely enforce linear independence among the features – with intuition as the basis. A much more methodological approach will be implemented further down the study. Table 1 shows the attributes that were chosen per category.

Data from the sixth survey was used for the targets. Mental health information for K6, PHQ-9, GAD-7, SSS-8, UCLA Loneliness Scale, LSNS-6, and SHS were gathered and bracketed according to their corresponding scales. Table 2 illustrates these scales.

2.3 ML Models

The Multilayer Perceptron (MLP) is a type of neural network that maps input data to predictions through multiple fully connected layers with neurons using activation functions (Fuchs, 2021). It uses a non-linear transformation like relu to capture complex patterns and is optimized using the adam solver. Key hyperparameters like hidden_layer_sizes control network depth, alpha applies regularization to prevent overfitting, and learning_rate_init adjusts the training speed (Scikit-Learn, n.d.).

Support Vector Machines (SVM) classify data by finding an optimal hyperplane that maximizes the margin between classes. The hyperplane is determined by support vectors, which are the closest data points to the decision boundary (Datacamp, n.d.). Key hyperparameters include C, which balances margin size and classification error, and kernel, which transforms data for non-linear separability (e.g., using the rbf kernel). Other hyperparameters like gamma, degree,

and class_weight help adjust model complexity, while probability enables probability estimates (Scikit-Learn, n.d.).

A Decision Tree Classifier splits data based on features to classify it, using a tree-like structure that starts at the root node and splits into branches based on criteria like Gini impurity. The process stops at leaf nodes, which represent final classifications (GeeksforGeeks, 2024). Key hyperparameters like max_depth and min_samples_split control the tree's depth and complexity to prevent overfitting, while options like class_weight help handle class imbalances (Scikit-Learn, n.d.).

Random Forest Classification is an ensemble method that improves accuracy by combining multiple decision trees, each trained on a random subset of data and features. Trees vote for a class, and the class with the majority vote is chosen (GeeksforGeeks, 2020). Key hyperparameters include n_estimators for the number of trees, max_depth to control tree depth and prevent overfitting, and min_samples_split to define the minimum samples needed to split a node. Other parameters like max_features increase randomness for better generalization, while oob_score allows model evaluation without a validation set (Scikit-Learn, n.d.).

The k-Nearest Neighbors (kNN) algorithm classifies a new data point by looking at the majority class among its k closest neighbors, determined by a chosen distance metric like Euclidean distance. It's an instance-based, distance-driven model that uses a voting system where nearby neighbors influence the classification of the new point (Datacamp, n.d.). Key hyperparameters, like n_neighbors to control k and weights to adjust neighbor influence (Scikit-Learn, n.d.).

Gaussian Naive Bayes (GNB) classifies data by assuming features are conditionally independent and normally distributed within each class. It calculates each feature's mean and variance per class to model them as Gaussian distributions. For a new data point, GNB uses these distributions to estimate class probabilities and selects the class with the highest likelihood (GeeksforGeeks, 2023). Key hyperparameters, like priors for handling class imbalance and var_smoothing to stabilize calculations with low variance, help improve accuracy and robustness, especially in high-dimensional applications like spam detection and text classification (Scikit-Learn, n.d.).

2.4 ML (train/test split, hyperparameter optimization, cross-validation, testing)

This section has not yet been completed, but partial results are in the GDrive.

3. RESULTS AND DISCUSSION

Descriptive results that show the distributions in each of the important metrics. Classification results that are tabulated. Explainability results that are tabulated. Feedback from domain experts that are tabulated.

4. CONCLUSION

A paragraph about the explainability results and how it plays a role in the generated classifications.

We expect to see generated counterfactual explanations that help in illuminating the details behind AI-generated post-COVID psychological profile predictions.

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